

Pressure Sensor Mat to Aid Repositioning of ICU Patients

Supervisor: Dr. Alexander Lunt

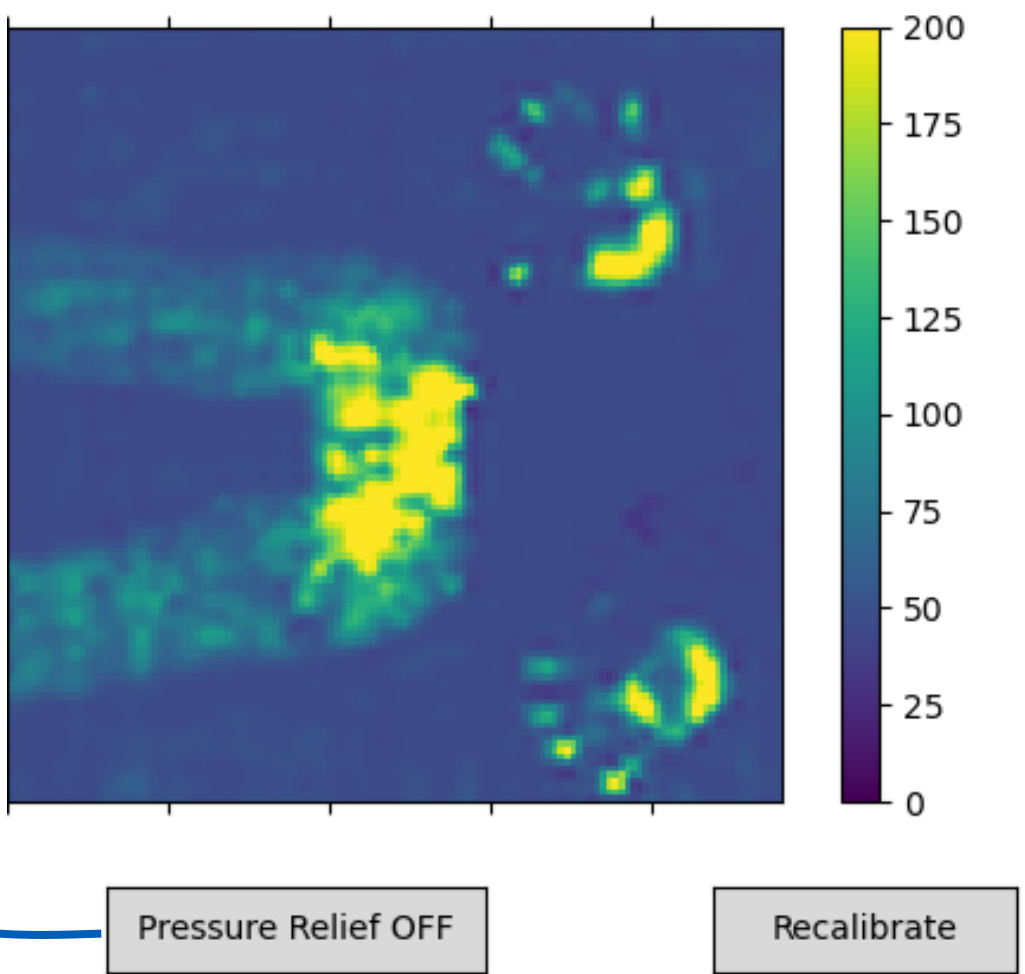
Assessor: Dr. Ioannis Georgilas

£1.5bn is spent annually on treating pressure sores

Proning can lead to a **45%** reduction in mortality for severe cases of ARDS

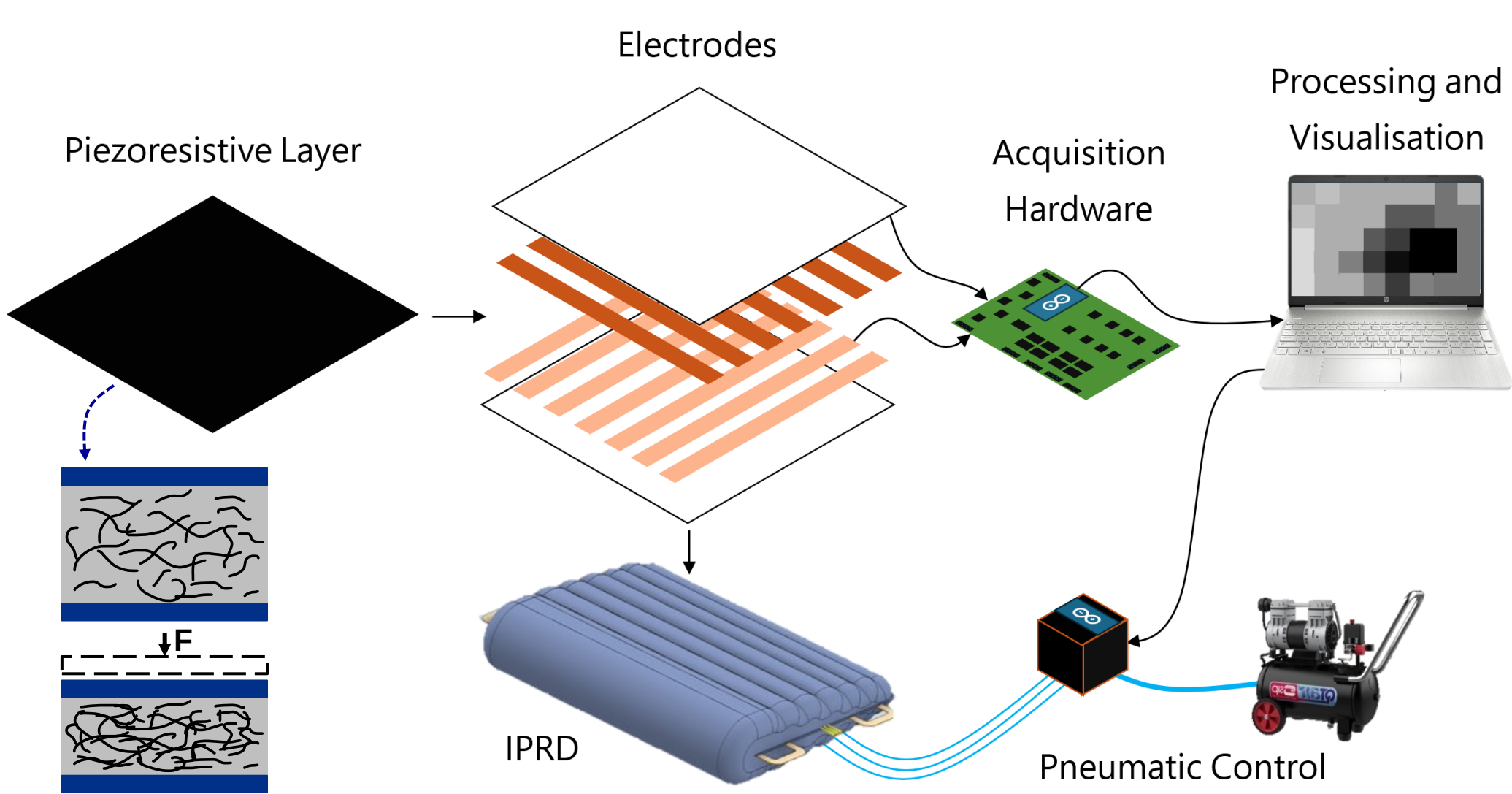
The **Inflatable Prone Repositioning Device** has been developed in to allow more frequent and less labour-intensive repositioning of prone ICU patients suffering from Acute Respiratory Distress Syndrome. **Real-time pressure distribution** data has been implemented to allow more effective manual repositioning, and to demonstrate closed-loop control to **automatically reposition patients** based on live pressure distribution data.

Live Pressure Distribution Data



Pressure Relief Mode:
Automatically redistributes pressure in the inflatable based on real-time pressure distribution data

Interactive User Interface



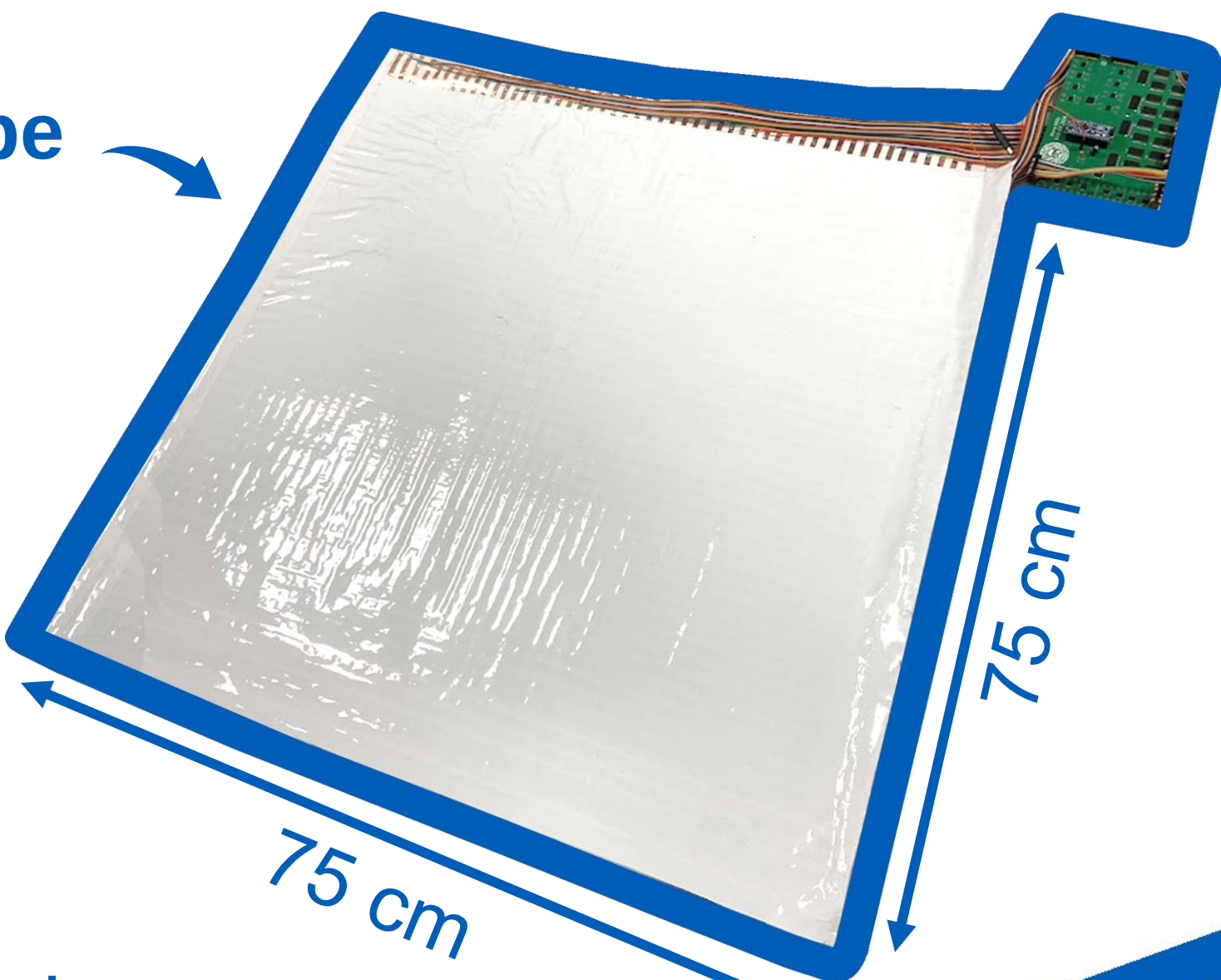
The sensor is based on **piezoresistive** materials which change electrical pressure depending on the mechanical force applied. Rows of electrodes are placed either side of the piezoresistive layer to create a matrix of sensing points.

Automatic control of the pneumatics was implemented using live pressure sensor data to inflate different parts of the device where pressure was highest.

Full scale prototype

Total System Cost: £250

- Arduino Nano**
ATMega 328P
- Fully Flexible**
2 mm thick
- 2 Hz refresh rate**
10 kHz voltage readings
- 4096 sensing points**
2 cm² cell size
- Automatic Repositioning**
Closed-loop control using real-time pressure data



PCB Design

A custom PCB was produced to drive the pressure sensor mat. Powered by an Arduino Nano, this implements a zero-potential scanning method (below) via analog switches to eliminate cross-talk effects caused by parasitic current paths.

Final PCB: 170x100 mm, £110, accommodates up to 64 x 64 arrays

